

Unsupervised Learning -- Programming Assignment

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Dataset/files to be used

- NYC Yellow Taxi data from September 2025 (yellow_tripdata_2025-09.zip)
- Data dictionary (Yellow_Taxi_Trip_Data_Data_Dictionary.xlsx)
- Zones (taxi_zone_lookup.csv)

Background Description

The New York City Taxi and Limousine Commission (TLC), created in 1971, is the agency responsible for licensing and regulating New York City's medallion (yellow) taxis, street hail livery (green) taxis, for-hire vehicles (FHV's), commuter vans, and paratransit vehicles. The TLC collects trip record information for each taxi and for-hire vehicle trip completed by licensed drivers and vehicles.

NYC Taxi

The records in the dataset are generated from the trip record submissions made by yellow taxi Technology Service Providers (TSPs). Each row represents a single trip in a yellow taxi. The trip records include fields capturing pick-up and drop-off dates/times, pick-up and drop-off taxi zone locations, trip distances, itemized fares, rate types, payment types, and driver-reported passenger counts

Task A: Perform clustering using the following features: 1. Distance 2. Fare 3. Tip 4. Rate-code 5. Pickup zone

Which distance metric works best for each feature?

Task B: From your submission on supervised learning, pick the *best* performing model. Using the same algorithm and best combination of hyperparameters from the previous submission, perform supervised learning on *this* dataset, and try to predict the `payment_type` label.

Task C: After clustering on various features in *Task A*, pick the best clusters and map cluster centroids back to the `payment_type` label. Does clustering outperform your best performing model from *Task B*?

Submission and Grading

Submission: Do this assignment in a notebook called `clustering_taxi.ipynb`.

Deadline: 27-November-2025 (this can be superseded by announcements or new deadline on Brightspace by the module coordinator)

You should discuss why (or why not) you perform each aspect of learning. The discussion is as important as the code, and lack of discussion will be graded *downwards* appropriately.
